

Paradigm Comparison for Clinical EHR Question-Answering

§1. Setup & methodology

Five retrieval paradigms answered each of 334 questions across three models (Claude Haiku 4.5, Qwen 2.5 72B, Llama 3.1 8B) and three patient-cohort tiers (200, 2 000, 20 000). Each (question, model, tier, system) cell received a 3-level score (0, 0.5, 1) from a hand-reasoned LLM-as-judge applying the rubric in docs/judge-calibration.md (17 adjudicated calibration cases). Final dataset: 13,640 judged cells.

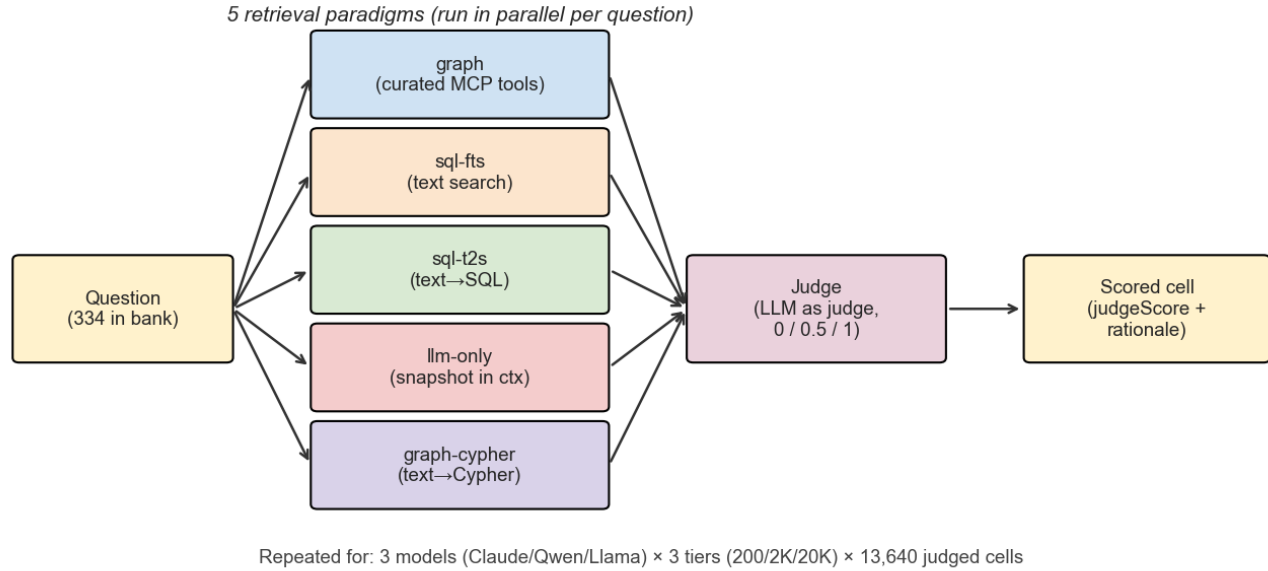


Figure 1: pipeline

Cells per (model × tier):

Model	tier-200	tier-2K	tier-20K
claude-haiku-4-5	1,479	1,599	1,470
llama-3.1-8b	1,600	1,585	1,416
qwen-2.5-72b	1,530	1,542	1,419

§2. Top-line accuracy

Pooled judge-mean across all 5 systems and all 6 question types, per (model × tier):

Model	Tier	n	Judge mean	% full	% partial	% wrong
claude-haiku-4-5	200	1,479	0.736	65.7%	15.8%	18.5%
claude-haiku-4-5	2000	1,599	0.690	60.2%	17.7%	22.1%
claude-haiku-4-5	20000	1,470	0.715	63.8%	15.4%	20.7%
llama-3.1-8b	200	1,600	0.304	23.9%	13.0%	63.1%
llama-3.1-8b	2000	1,585	0.366	28.1%	17.1%	54.8%
llama-3.1-8b	20000	1,416	0.314	23.4%	15.8%	60.7%
qwen-2.5-72b	200	1,530	0.600	51.0%	18.1%	30.9%
qwen-2.5-72b	2000	1,542	0.619	53.2%	17.4%	29.4%
qwen-2.5-72b	20000	1,419	0.655	56.9%	17.1%	25.9%

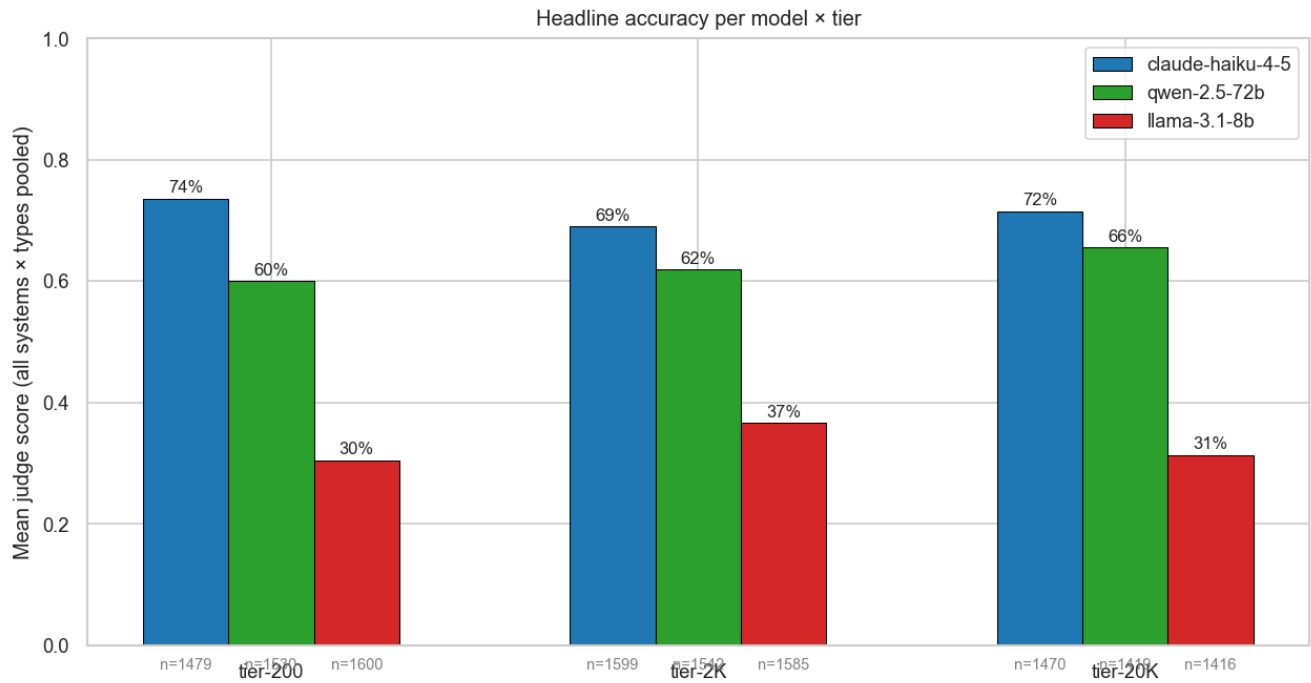


Figure 2: headline

Interpretation: Claude leads at every tier; Qwen ranks second; Llama trails by a wide margin under the pooled view because tool-using systems contribute many zeros for that model.

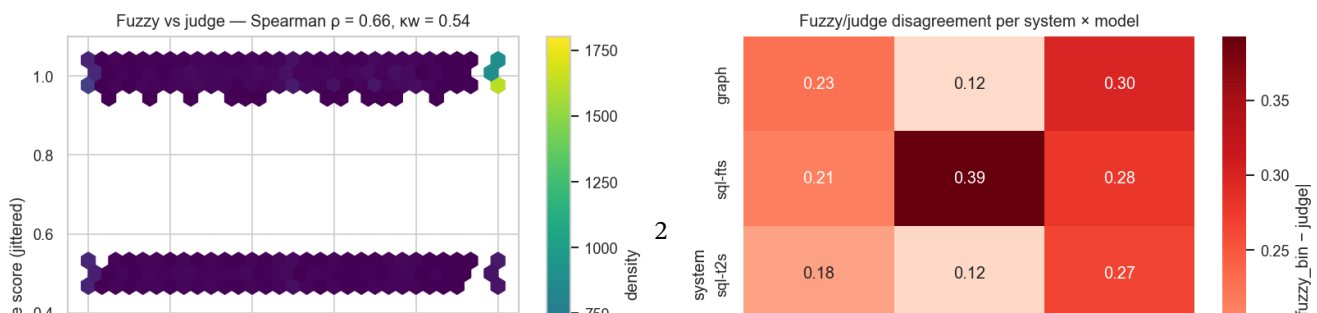
Best paradigm per (model x tier) — the practical view if you deploy only the strongest system for each model:

Model	Tier	Best system	Best judge mean	Pooled mean (all 5 systems)
claude-haiku-4-5	200	sql-t2s	0.885	0.736
claude-haiku-4-5	2000	sql-t2s	0.806	0.690
claude-haiku-4-5	20000	graph-cypher	0.837	0.715
llama-3.1-8b	200	sql-fts	0.676	0.304
llama-3.1-8b	2000	sql-fts	0.646	0.366
llama-3.1-8b	20000	sql-fts	0.592	0.314
qwen-2.5-72b	200	sql-t2s	0.736	0.600
qwen-2.5-72b	2000	sql-t2s	0.730	0.619
qwen-2.5-72b	20000	sql-t2s	0.766	0.655

Interpretation: once each model picks its best paradigm, the headline gap narrows but remains large: Claude 0.85, Qwen 0.75, Llama 0.55–0.62 (sql-fts wins for Llama because of its tool-calling failure mode).

§3. Metric validity: fuzzy vs hand-judge

The original automated metric is a fuzzy token-overlap score against the ground-truth string. The LLM judge applies a hand-reasoned clinical rubric. The judge is the metric used everywhere else in this report; this section documents agreement between the two.



Interpretation: moderate agreement only ($\kappa \approx 0.44\text{--}0.52$). The fuzzy scorer systematically penalises long-form clinical answers and fails to recognise correct UNANSWERABLE refusals, so we use the judge throughout this report.

§4. Paradigm ranking per (model $\times$ tier)

System means with 95% bootstrap CIs:

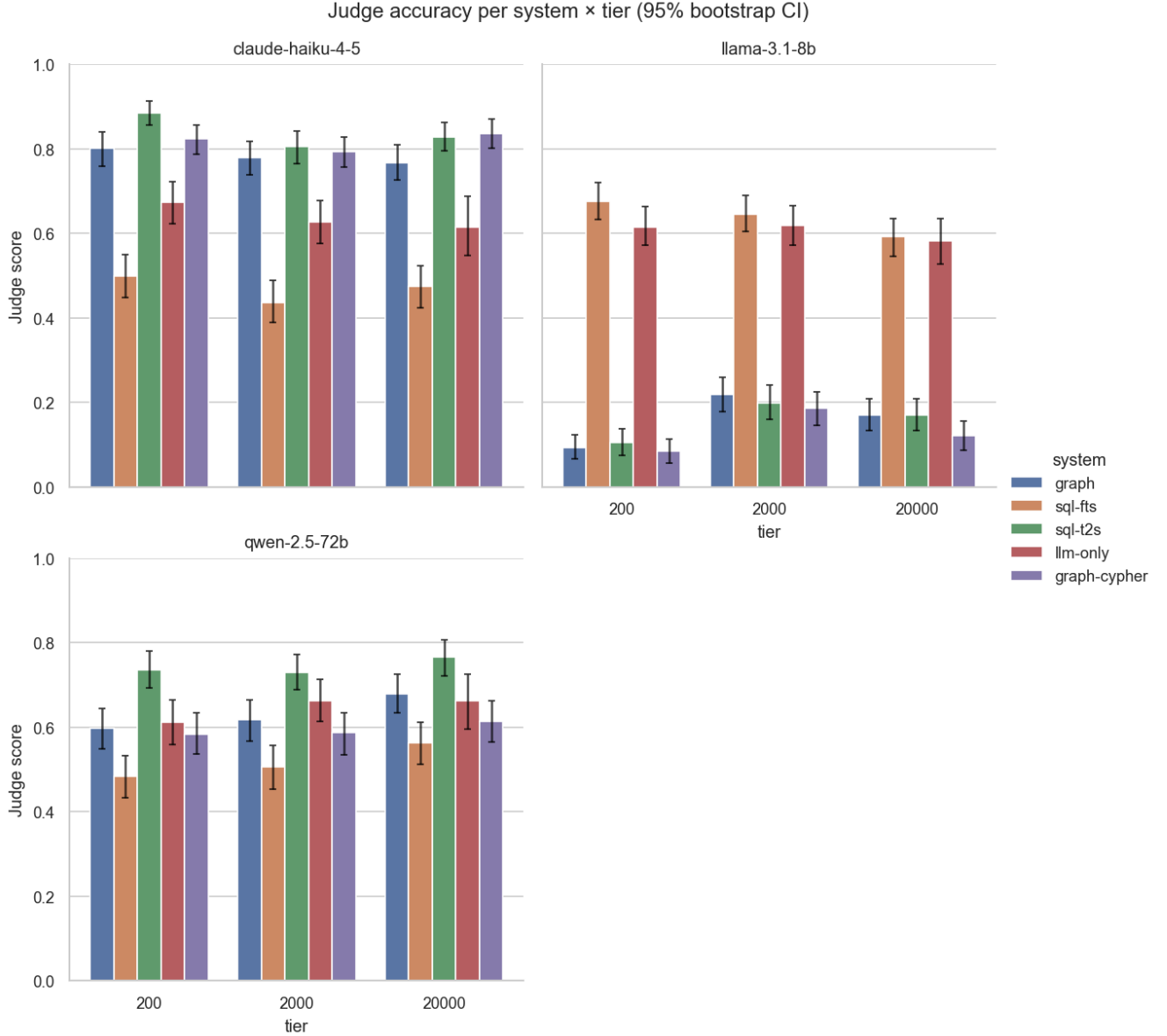


Figure 4: system means

Failure-mode breakdown — fraction of cells scoring 1.0, 0.5, and 0:

Model	Tier	System	n	Mean	95% CI
claude-haiku-4-5	200	graph	281	0.801	[0.760, 0.840]
claude-haiku-4-5	200	graph-cypher	330	0.824	[0.788, 0.856]

Model	Tier	System	n	Mean	95% CI
claude-haiku-4-5	200	llm-only	262	0.674	[0.622, 0.721]
claude-haiku-4-5	200	sql-fts	315	0.500	[0.448, 0.551]
claude-haiku-4-5	200	sql-t2s	291	0.885	[0.856, 0.912]
claude-haiku-4-5	2000	graph	326	0.779	[0.739, 0.817]
claude-haiku-4-5	2000	graph-cypher	332	0.794	[0.757, 0.828]
claude-haiku-4-5	2000	llm-only	274	0.628	[0.577, 0.677]
claude-haiku-4-5	2000	sql-fts	334	0.436	[0.389, 0.488]
claude-haiku-4-5	2000	sql-t2s	333	0.806	[0.766, 0.842]
claude-haiku-4-5	20000	graph	327	0.768	[0.726, 0.809]
claude-haiku-4-5	20000	graph-cypher	331	0.837	[0.802, 0.870]
claude-haiku-4-5	20000	llm-only	147	0.616	[0.548, 0.687]
claude-haiku-4-5	20000	sql-fts	333	0.474	[0.423, 0.523]
claude-haiku-4-5	20000	sql-t2s	332	0.828	[0.795, 0.861]
llama-3.1-8b	200	graph	329	0.093	[0.067, 0.123]
llama-3.1-8b	200	graph-cypher	331	0.085	[0.057, 0.113]
llama-3.1-8b	200	llm-only	274	0.615	[0.571, 0.664]
llama-3.1-8b	200	sql-fts	333	0.676	[0.632, 0.721]
llama-3.1-8b	200	sql-t2s	333	0.105	[0.075, 0.138]
llama-3.1-8b	2000	graph	325	0.218	[0.178, 0.260]
llama-3.1-8b	2000	graph-cypher	323	0.186	[0.146, 0.226]
llama-3.1-8b	2000	llm-only	272	0.619	[0.572, 0.665]
llama-3.1-8b	2000	sql-fts	333	0.646	[0.604, 0.689]
llama-3.1-8b	2000	sql-t2s	332	0.199	[0.160, 0.241]
llama-3.1-8b	20000	graph	306	0.170	[0.134, 0.209]
llama-3.1-8b	20000	graph-cypher	288	0.122	[0.087, 0.156]
llama-3.1-8b	20000	llm-only	188	0.582	[0.527, 0.636]
llama-3.1-8b	20000	sql-fts	331	0.592	[0.545, 0.636]
llama-3.1-8b	20000	sql-t2s	303	0.170	[0.134, 0.210]
qwen-2.5-72b	200	graph	328	0.598	[0.549, 0.645]
qwen-2.5-72b	200	graph-cypher	293	0.584	[0.536, 0.633]
qwen-2.5-72b	200	llm-only	271	0.611	[0.559, 0.664]
qwen-2.5-72b	200	sql-fts	331	0.483	[0.432, 0.532]
qwen-2.5-72b	200	sql-t2s	307	0.736	[0.692, 0.780]
qwen-2.5-72b	2000	graph	333	0.617	[0.566, 0.665]
qwen-2.5-72b	2000	graph-cypher	310	0.587	[0.535, 0.634]
qwen-2.5-72b	2000	llm-only	261	0.663	[0.613, 0.713]
qwen-2.5-72b	2000	sql-fts	321	0.506	[0.453, 0.556]
qwen-2.5-72b	2000	sql-t2s	317	0.730	[0.689, 0.771]
qwen-2.5-72b	20000	graph	327	0.679	[0.635, 0.725]
qwen-2.5-72b	20000	graph-cypher	297	0.613	[0.564, 0.663]
qwen-2.5-72b	20000	llm-only	167	0.662	[0.596, 0.725]
qwen-2.5-72b	20000	sql-fts	323	0.562	[0.512, 0.611]

Model	Tier	System	n	Mean	95% CI
qwen-2.5-72b	20000	sql-t2s	305	0.766	[0.721, 0.807]

Interpretation: for Claude and Qwen, sql-t2s consistently ranks first and sql-fts last; graph and graph-cypher are statistically indistinguishable from each other. For Llama, the ordering flips entirely: sql-fts and llm-only are best, the three tool-using systems collapse to 10–20%.

§5. Statistical confirmation of the ranking

Four tests, sequenced omnibus [7] pairwise, with each available in both ordinal and binary form.

§5.1 Friedman (ordinal omnibus)

H_0 : all five systems have the same distribution of judge scores on the same questions.

Model	Tier	n questions	χ^2	p
claude-haiku-4-5	200	226	104.8	9.22e-22
claude-haiku-4-5	2000	267	162.6	4.02e-34
claude-haiku-4-5	20000	143	86.7	6.60e-18
llama-3.1-8b	200	266	589.1	3.59e-126
llama-3.1-8b	2000	255	363.7	1.89e-77
llama-3.1-8b	20000	160	266.6	1.72e-56
qwen-2.5-72b	200	214	35.1	4.42e-07
qwen-2.5-72b	2000	227	25.6	3.76e-05
qwen-2.5-72b	20000	145	19.1	7.66e-04

Interpretation: every combination rejects H_0 at $p \leq 0.001$ — at least one system differs.

§5.2 Cochran’s Q (binary omnibus)

Same hypothesis as Friedman but with the judge score binarised. Strict cutoff = judge == 1.0; lenient cutoff = judge ≥ 0.5 .

Cutoff	Model	Tier	n	k	Q	p
strict	claude-haiku-4-5	200	226	5	82.2	6.06e-17
strict	claude-haiku-4-5	2000	267	5	137.2	1.14e-28
strict	claude-haiku-4-5	20000	143	5	74.5	2.57e-15
strict	llama-3.1-8b	200	266	5	388.8	7.48e-83
strict	llama-3.1-8b	2000	255	5	198.7	7.31e-42
strict	llama-3.1-8b	20000	160	5	134.1	5.24e-28
strict	qwen-2.5-72b	200	214	5	31.7	2.25e-06
strict	qwen-2.5-72b	2000	227	5	16.6	2.32e-03
strict	qwen-2.5-72b	20000	145	5	23.9	8.51e-05
lenient	claude-haiku-4-5	200	226	5	137.0	1.23e-28
lenient	claude-haiku-4-5	2000	267	5	150.0	2.07e-31
lenient	claude-haiku-4-5	20000	143	5	102.4	3.01e-21
lenient	llama-3.1-8b	200	266	5	583.6	5.62e-125
lenient	llama-3.1-8b	2000	255	5	366.3	5.18e-78
lenient	llama-3.1-8b	20000	160	5	287.4	5.55e-61
lenient	qwen-2.5-72b	200	214	5	26.7	2.23e-05
lenient	qwen-2.5-72b	2000	227	5	37.3	1.56e-07
lenient	qwen-2.5-72b	20000	145	5	15.7	3.41e-03

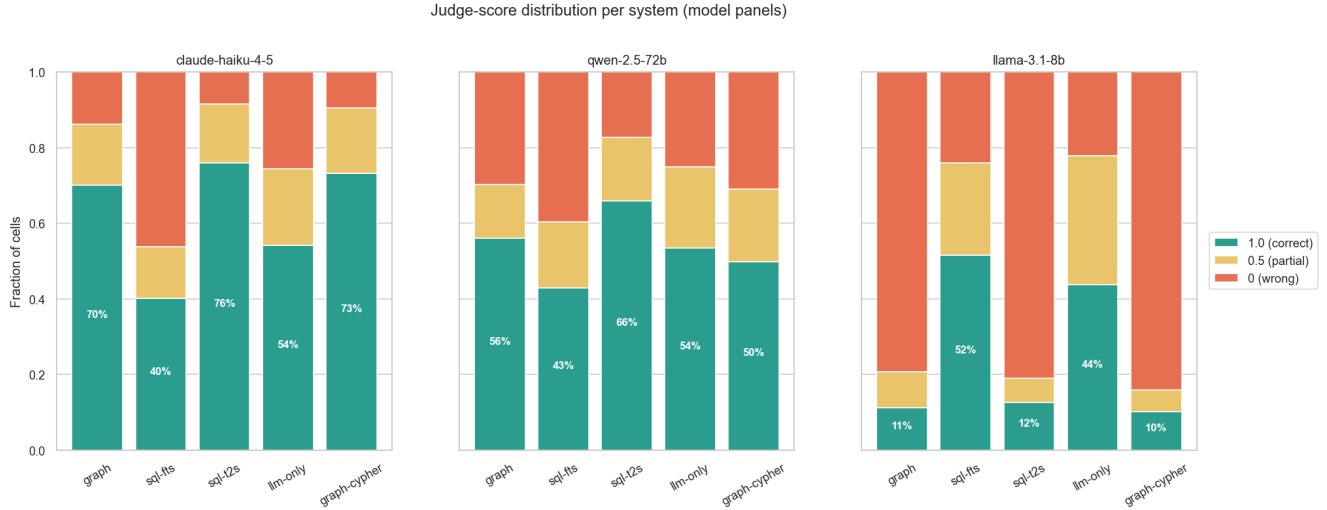


Figure 5: score distribution

Interpretation: Cochran’s Q rejects H_0 at $p \leq 0.001$ in all 18 combinations (9×2 cutoffs) — the binary framing confirms the ordinal one.

§5.3 Pairwise Wilcoxon signed-rank (ordinal, Bonferroni-corrected)

Each cell shows mean(row) – mean(col). Stars: * $p < .05$, ** $p < .01$, *** $p < .001$.

Full table with Cliff’s δ effect sizes: `tables/3_pairwise_wilcoxon.csv`.

Interpretation: for Claude and Qwen, sql-t2s significantly beats every other system in nearly every (tier) combination. For Llama, sql-fts and llm-only significantly beat the three tool-using systems by 30–55 percentage points.

§5.4 Pairwise McNemar (binary, Bonferroni-corrected)

Strict cutoff:

Lenient cutoff:

Full McNemar table with discordant-pair counts (b/c): `tables/3c_mcnemar_pairs.csv`.

Interpretation: McNemar agrees with Wilcoxon — the same pairs are significant under both cutoffs, with effect sizes 5–55 percentage points.

§6. Where each paradigm wins (question type $\times$ system)

Mean judge score per (model $\times$ type $\times$ system), pooled across tiers:

Best paradigm per (model $\times$ question type) — derived from the heatmap:

Type	Claude best (score)	Qwen best (score)	Llama best (score)
simple-lookup	sql-t2s (0.91)	sql-t2s (0.88)	sql-fts (0.84)
cohort	sql-t2s (0.77)	graph (0.75)	sql-fts (0.40)
temporal	graph-cypher (0.82)	sql-t2s (0.73)	sql-fts (0.68)
multi-hop	sql-t2s (0.94)	sql-t2s (0.66)	sql-fts (0.55)
reasoning	sql-t2s (0.80)	sql-t2s (0.75)	sql-fts (0.69)
unanswerable	sql-fts (0.74)	graph (0.74)	llm-only (0.59)

Interpretation: sql-t2s wins the most type slots for capable models; graph wins specifically on cohort. For Llama every type’s winner is sql-fts or llm-only (i.e. the non-tool-using paths).

Pairwise system differences (judge), Bonferroni-adjusted: *p<.05, **p<.01, ***p<.001

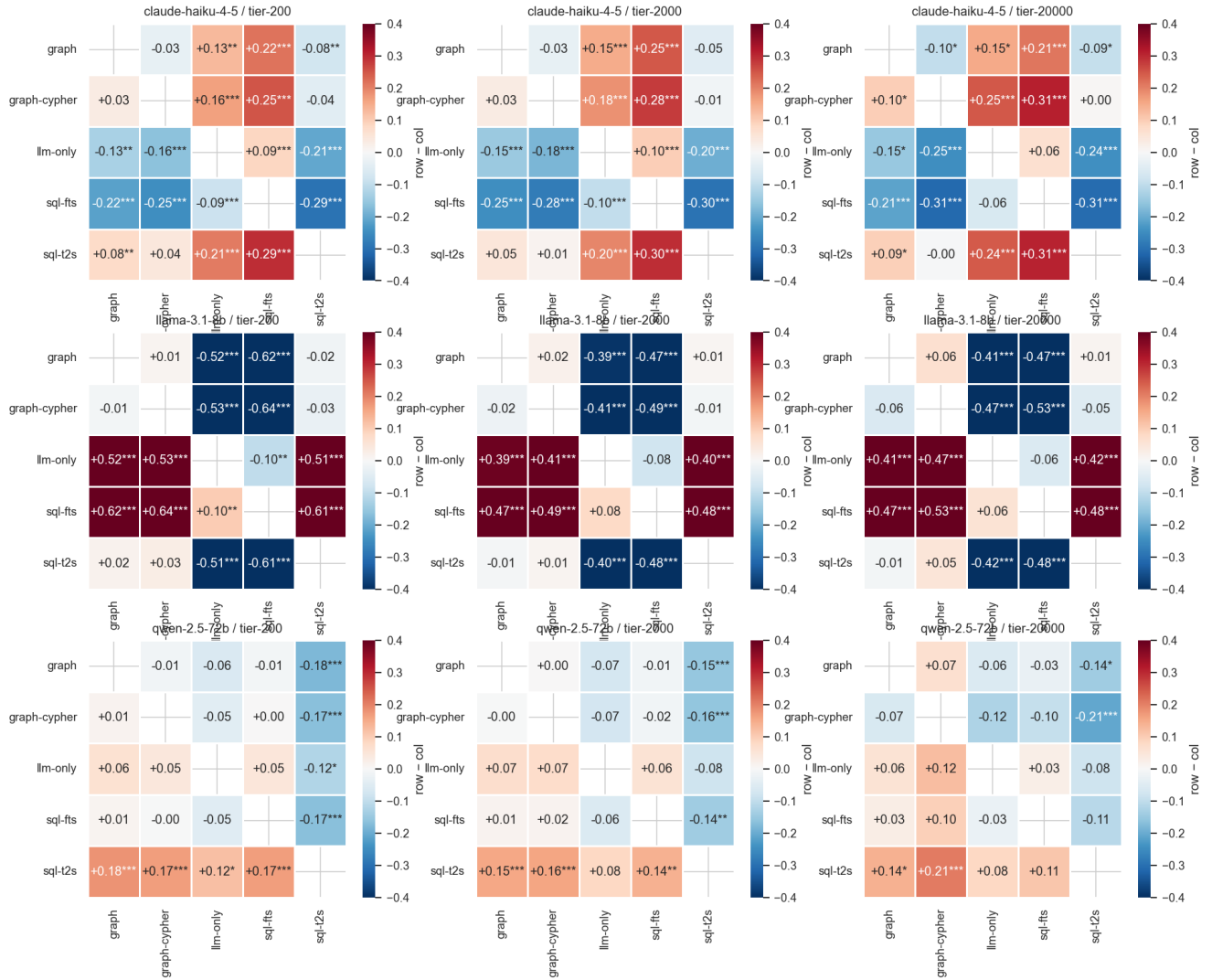


Figure 6: pairwise heatmaps

McNemar pairwise diffs (strict cutoff), Bonferroni-adjusted: *p<.05, **p<.01, ***p<.001

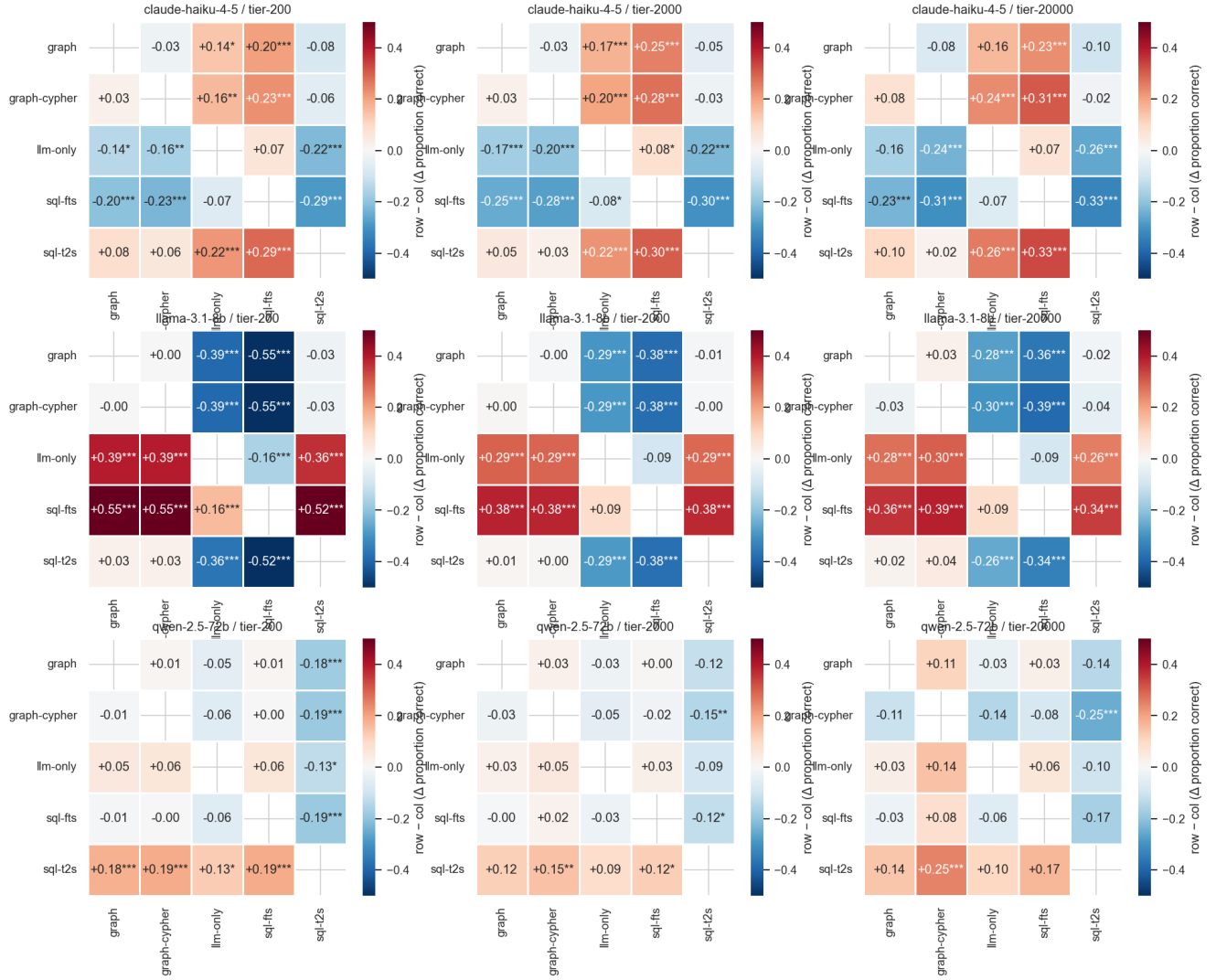


Figure 7: mcnemar strict

McNemar pairwise diffs (lenient cutoff), Bonferroni-adjusted: *p<.05, **p<.01, ***p<.001

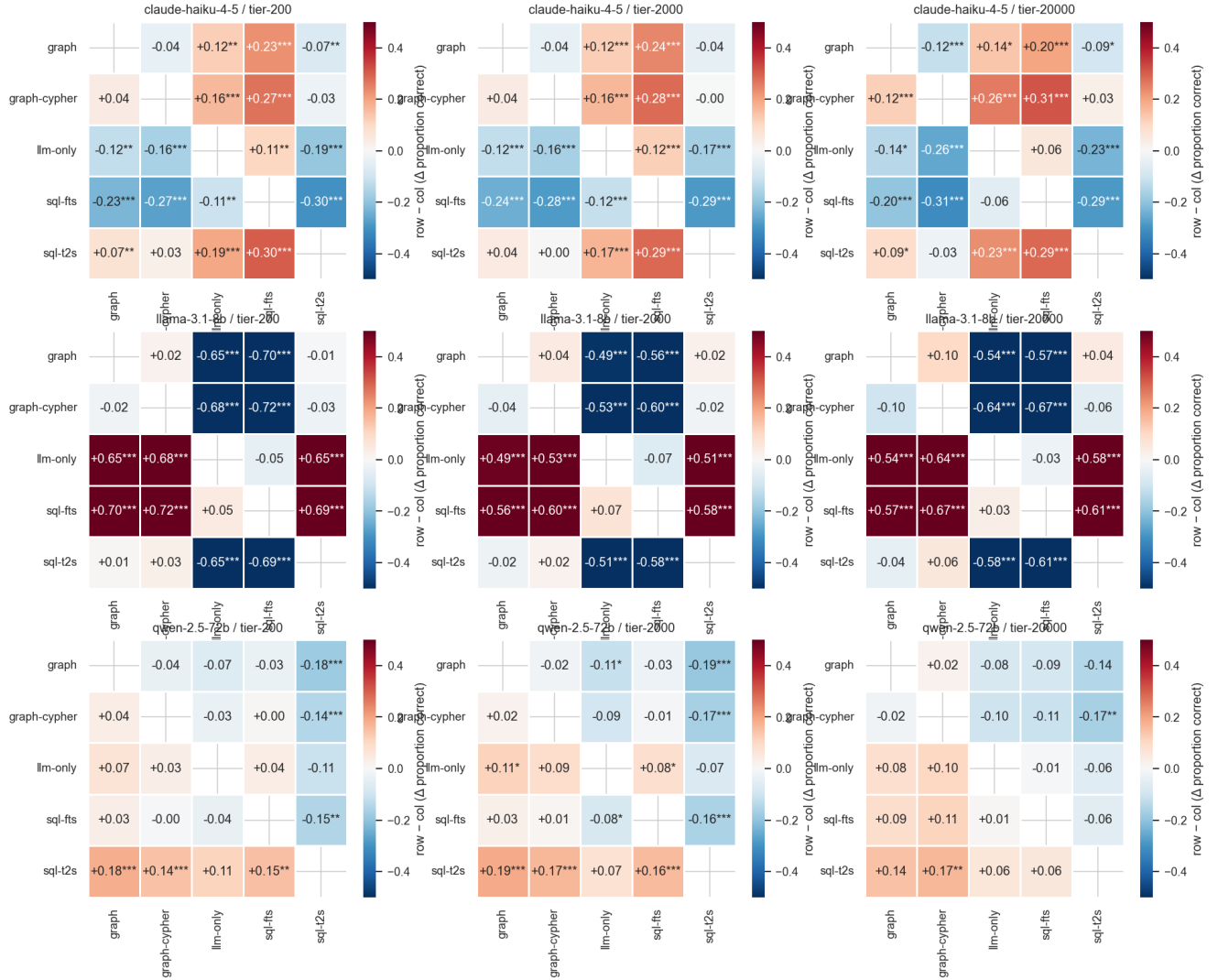


Figure 8: mcnemar lenient

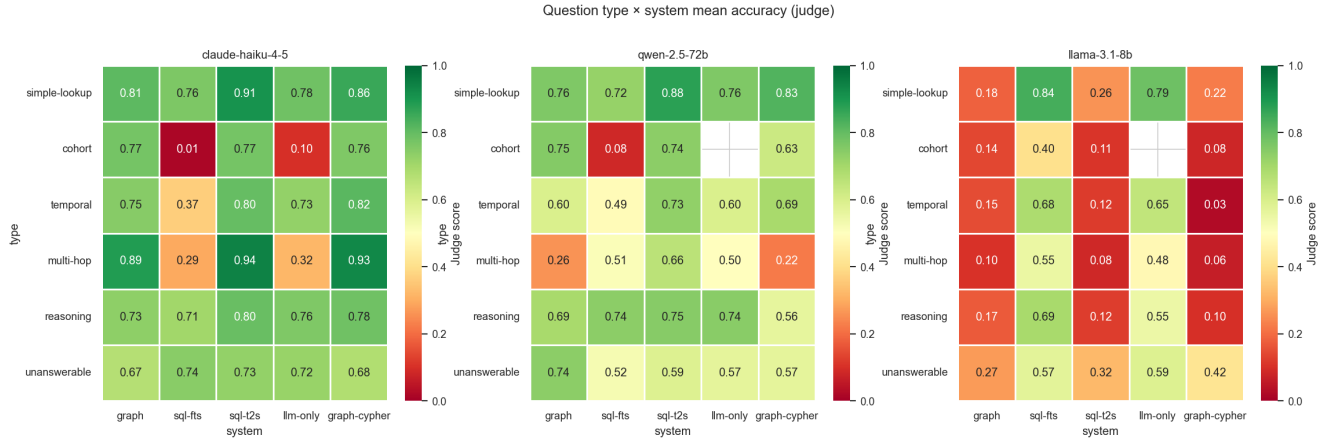


Figure 9: type-system heatmap

§7. Scaling across cohort sizes

Mean judge score per (model × system) across the three patient-cohort tiers:

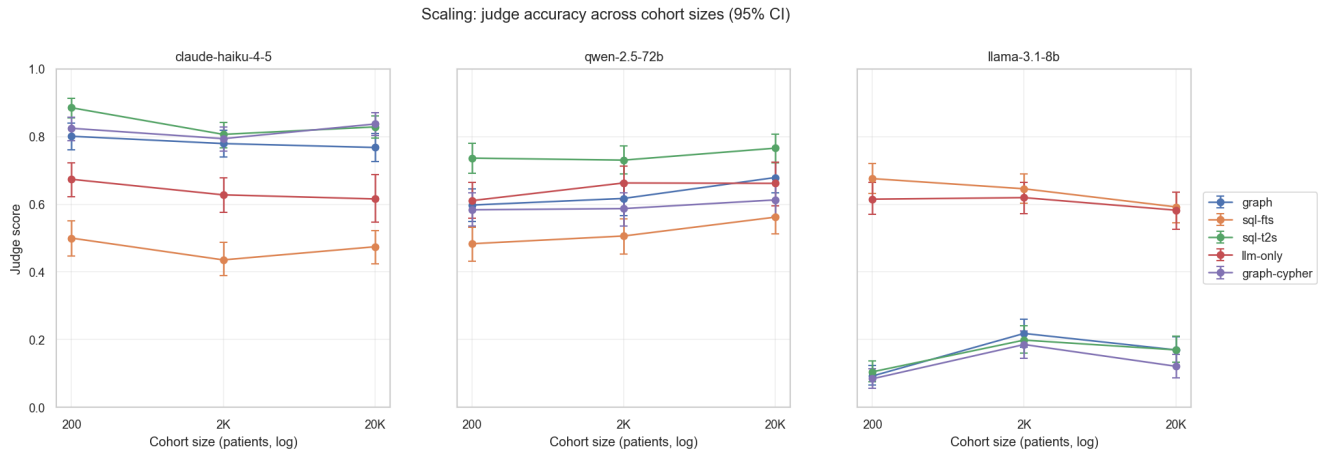


Figure 10: scaling curves

Linear regression of judge on $\log_{10}(\text{tier})$:

Model	System	Slope per $\log_{10}$ tier	95% CI	r	p
claude-haiku-4-5	graph	-0.016	[-0.045, +0.012]	-0.04	2.64e-01
claude-haiku-4-5	graph-cypher	+0.006	[-0.019, +0.031]	+0.02	6.17e-01
claude-haiku-4-5	llm-only	-0.031	[-0.073, +0.011]	-0.06	1.48e-01
claude-haiku-4-5	sql-fts	-0.012	[-0.048, +0.024]	-0.02	5.01e-01
claude-haiku-4-5	sql-t2s	-0.027	[-0.052, -0.003]	-0.07	2.96e-02
llama-3.1-8b	graph	+0.040	[+0.014, +0.066]	+0.10	2.63e-03
llama-3.1-8b	graph-cypher	+0.020	[-0.004, +0.045]	+0.05	1.07e-01

Model	System	Slope per $\log_{10}$ tier	95% CI	r	p
llama-3.1-8b	llm-only	-0.015	[-0.051, +0.021]	-0.03	4.22e-01
llama-3.1-8b	sql-fts	-0.042	[-0.073, -0.010]	-0.08	9.00e-03
llama-3.1-8b	sql-t2s	+0.033	[+0.007, +0.060]	+0.08	1.36e-02
qwen-2.5-72b	graph	+0.041	[+0.007, +0.075]	+0.07	1.91e-02
qwen-2.5-72b	graph-cypher	+0.015	[-0.021, +0.050]	+0.03	4.20e-01
qwen-2.5-72b	llm-only	+0.028	[-0.012, +0.068]	+0.05	1.72e-01
qwen-2.5-72b	sql-fts	+0.039	[+0.004, +0.074]	+0.07	2.72e-02
qwen-2.5-72b	sql-t2s	+0.015	[-0.016, +0.045]	+0.03	3.46e-01

Interpretation: slopes are small ($|\text{slope}| < 0.05$ in every case); CIs for most slopes cross zero. No paradigm shows pronounced accuracy degradation from 200 to 20 000 patients.

§8. Cross-model comparison on shared questions

Tool-using mean – llm-only mean, per (model $\times$ tier):

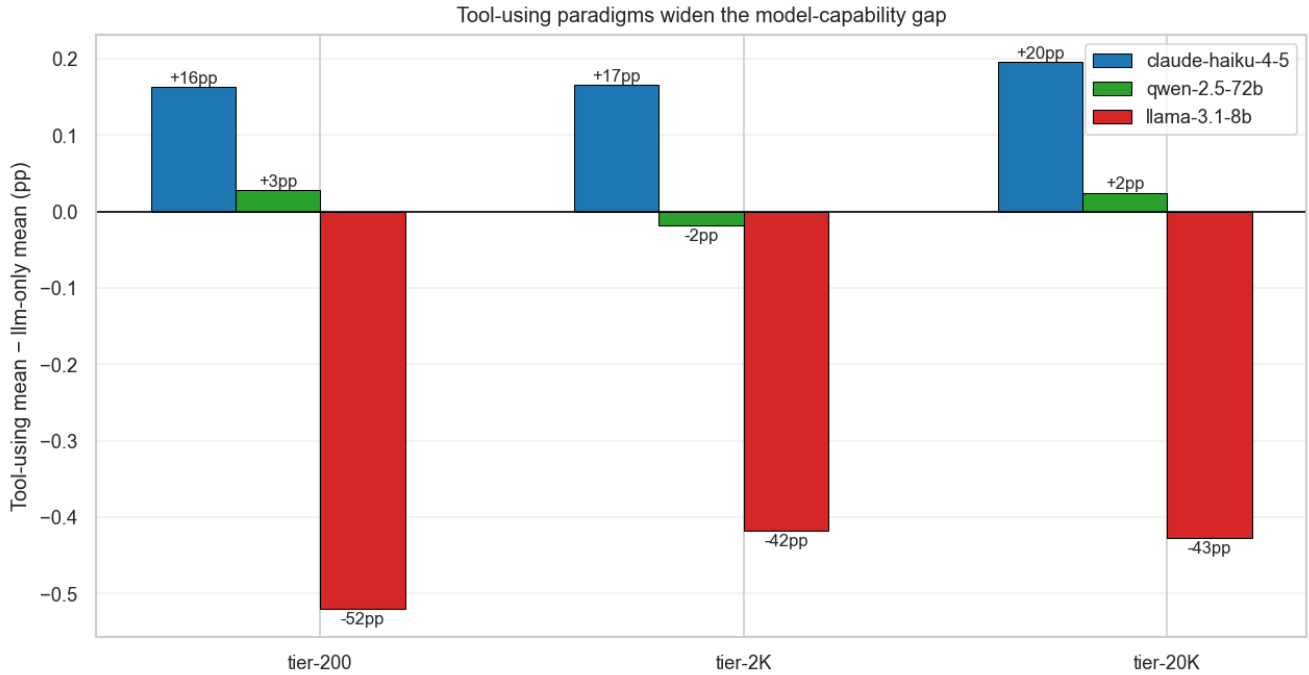


Figure 11: tool vs llm-only gap

Win rate — fraction of questions where each system is best-or-tied:

Friedman test on shared questions (Claude vs Qwen vs Llama, paired per question):

Which system wins each question? (per model × tier)

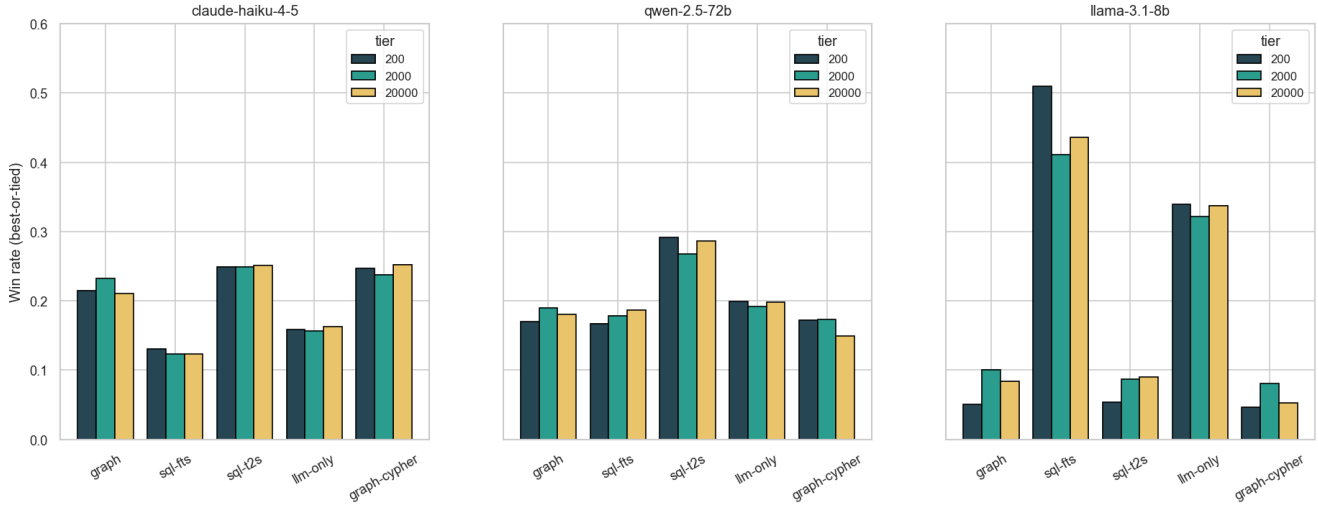


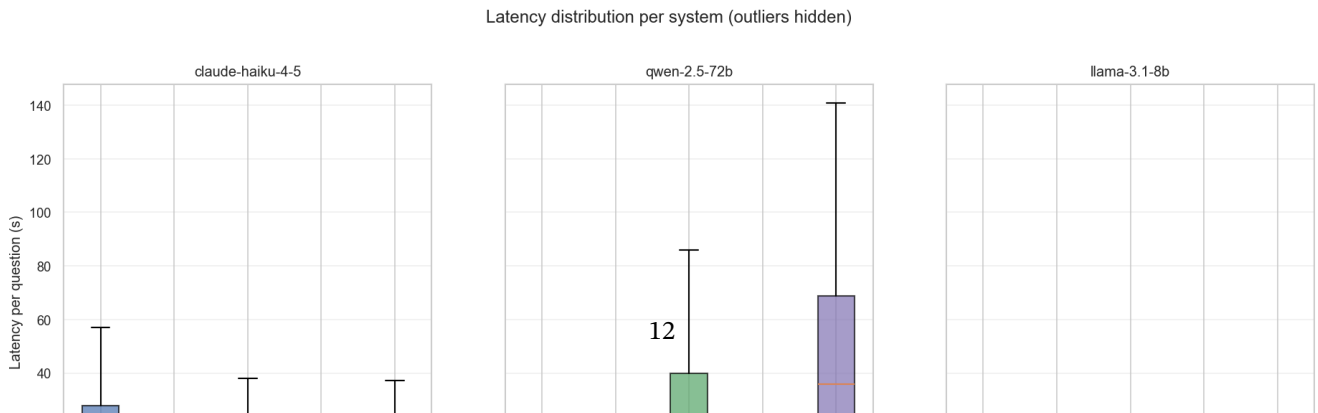
Figure 12: win rate

System	Tier	n	Claude	Qwen	Llama	χ^2	p
graph	200	271	0.80	0.61	0.10	305.2	5.22e-67
graph	2000	316	0.78	0.61	0.22	214.5	2.61e-47
graph	20000	295	0.76	0.69	0.17	269.1	3.65e-59
graph-cypher	200	288	0.82	0.59	0.09	315.9	2.52e-69
graph-cypher	2000	297	0.79	0.59	0.19	243.0	1.68e-53
graph-cypher	20000	260	0.82	0.63	0.12	276.0	1.18e-60
llm-only	200	249	0.69	0.61	0.61	12.5	1.96e-03
llm-only	2000	259	0.62	0.66	0.63	2.2	3.26e-01
llm-only	20000	129	0.62	0.70	0.56	12.9	1.58e-03
sql-fts	200	311	0.50	0.48	0.67	43.1	4.39e-10
sql-fts	2000	320	0.43	0.50	0.64	67.3	2.39e-15
sql-fts	20000	319	0.47	0.56	0.59	21.6	2.06e-05
sql-t2s	200	265	0.89	0.74	0.11	351.3	5.19e-77
sql-t2s	2000	315	0.81	0.73	0.20	305.8	3.88e-67
sql-t2s	20000	276	0.83	0.77	0.17	289.5	1.36e-63

Interpretation: tool-using paradigms give Claude +16 to +20pp over llm-only, give Qwen approximately zero net change, and cost Llama −42 to −52pp. The model-capability gap on the same questions is far larger for tool-using paradigms than for llm-only.

§9. Operational tradeoffs: latency and tool-loop length

Per-question latency in seconds (boxplots, outliers hidden):



Model	System	n	Median (s)	Mean (s)
claude-haiku-4-5	sql-fts	982	5.4	6.9
claude-haiku-4-5	sql-t2s	956	12.2	17.5
llama-3.1-8b	graph	960	3.7	5.7
llama-3.1-8b	graph-cypher	942	5.0	11.3
llama-3.1-8b	llm-only	734	1.7	3.3
llama-3.1-8b	sql-fts	997	1.2	3.1
llama-3.1-8b	sql-t2s	968	3.5	7.9
qwen-2.5-72b	graph	988	6.6	9.9
qwen-2.5-72b	graph-cypher	900	35.9	53.8
qwen-2.5-72b	llm-only	699	3.9	5.6
qwen-2.5-72b	sql-fts	975	2.1	3.6
qwen-2.5-72b	sql-t2s	929	19.2	30.8

Interpretation: sql-fts and llm-only are single-turn and fastest (~2–6 s median). Tool-using systems take several LLM²tool turns; Qwen graph-cypher runs longest (~36 s median) because of repeated query-fix loops.

§10. Error patterns and threats to validity

Two flags emitted alongside the judge score: `gt_suspect` (judge believes the ground truth is wrong) and `over_answered` (correct facts present but buried in extras).

Recurring GT issues flagged across multiple cells (top 15 by flag count):

QID	Type	Times flagged
UNA-108	unanswerable	15
RSN-76	reasoning	14
COH-24	cohort	13
RSN-56	reasoning	12
COH-29	cohort	11
RSN-54	reasoning	10
SL-14	simple-lookup	9
COH-31	cohort	9
TMP-12	temporal	8
COH-40	cohort	7
SL-160	simple-lookup	6
MH-53	multi-hop	6
COH-4	cohort	6
COH-37	cohort	4
SL-BRIDGE-4	simple-lookup	4

Interpretation: the most-flagged questions cluster in cohort (COH-23/24/26/27/28/29 — numeric mismatches), unanswerable (UNA-108 — income IS in Synthea), and reasoning (RSN-54/56/76 — over-narrow guideline GTs). These are dataset issues, not system failures.

§11. Files manifest

File	Description
tables/long_format.csv	Every judged cell — the master file for re-analysis
tables/1_descriptive_means.csv	System means + 95% bootstrap CIs
tables/2_friedman.csv	Friedman omnibus test per (model, tier)
tables/3_pairwise_wilcoxon.csv	Pairwise Wilcoxon + Cliff’s δ + Bonferroni p

File	Description
tables/3b_cochran_q.csv	Cochran’s Q omnibus (strict + lenient cutoffs)
tables/3c_mcnemar_pairs.csv	Pairwise McNemar with b/c counts
tables/4_scaling_regression.csv	log-tier regression slopes
tables/5_type_system_means.csv	Mean judge per (model, type, system)
tables/6_fuzzy_judge_agreement.csv	Spearman + weighted κ per model
tables/8_flag_rates.csv	gt_suspect + over_answered rates
tables/12_latency.csv	Median + mean latency per (model, system)
tables/9_cross_model_friedman.csv	Friedman across models per (system, tier)
tables/10_top_summary.csv	Top-line summary used in §2
tables/15_best_paradigm_per_model.csv	Best system per (model, tier) for §2
tables/16_best_system_per_type.csv	Best system per (model, type) for §6
tables/17_gt_flagged_questions.csv	GT-suspect counts per question

Schema of **long_format.csv**

Column	Type	Description
model	str	claude-haiku-4-5 / qwen-2.5-72b / llama-3.1-8b
tier	int	200 / 2000 / 20000 patient-cohort size
system	str	graph / sql-fts / sql-t2s / llm-only / graph-cypher
qid	str	Question identifier (e.g. SL-19, COH-14)
type	str	simple-lookup / cohort / temporal / multi-hop / reasoning / unanswerable
judge	float	0.0, 0.5, or 1.0 — primary outcome
fuzzy	float	Original token-overlap score; secondary outcome
latency_ms	float	End-to-end latency for the system’s answer
cost_usd	float	API cost (Claude only; OpenRouter omits)
turns	int	Number of LLM ² tool turns (tool-using systems only)
gt_suspect	bool	Judge flagged the ground-truth as suspect
over_answered	bool	Judge flagged answer as containing correct facts amid significant unrelated content

Tool-loop length per system (outliers hidden)

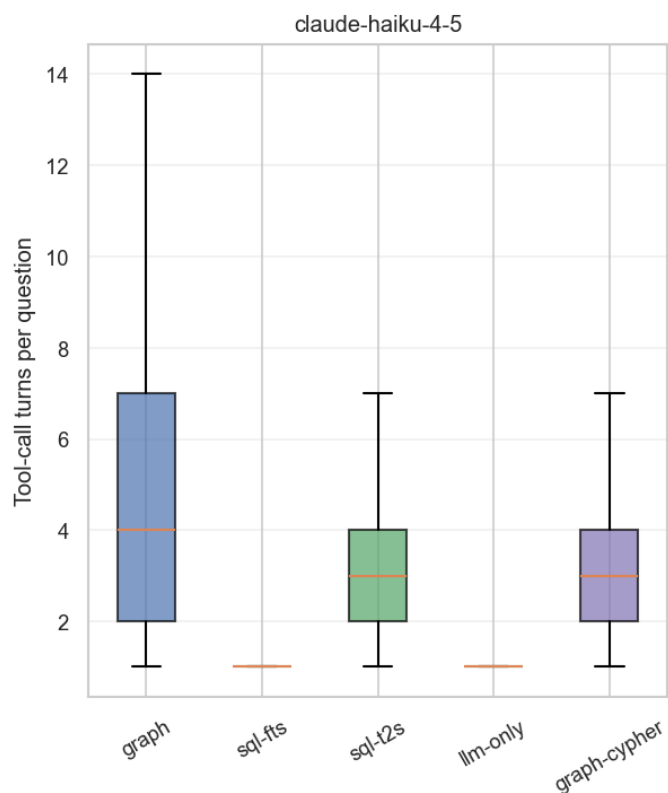


Figure 14: turns

Rate of gt_suspect=true			
	claude-haiku-4-5	llama-3.1-8b model	qwen-2.5-72b
graph	3.43%	1.25%	1.01%
sql-fts	2.04%	1.50%	0.82%
sql-t2s	4.08%	0.41%	0.97%
llm-only	2.64%	0.54%	0.29%
graph-cypher	3.93%	0.85%	1.00%

Rate of over_answered=true			
	claude-haiku-4-5	llama-3.1-8b model	qwen-2.5-72b
graph	2.68%	0.83%	0.61%
sql-fts	5.19%	1.71%	1.44%
sql-t2s	11.30%	1.24%	8.07%
llm-only	4.69%	3.00%	1.00%
graph-cypher	10.67%	2.12%	6.78%

Figure 15: flag rates